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Introduction:

In the sphere of mathematical modelling, two-compartment models serve as a cornerstone for the analysis and prediction of the behaviour of a wide range of systems, including but not limited to pharmacokinetics and environmental sciences. Conceptually, these models bifurcate a system into two interacting compartments, allowing for an in-depth study of the transfer of substances - such as drugs or environmental pollutants - between these areas over time. A pivotal aspect of these models is the estimation of transfer rates between the compartments, a process integral to understanding the system's mechanics and enhancing the accuracy of model predictions.

This paper delves into the advanced techniques of solving the system of eigenvalues and eigenvectors, a method critical in analysing the stability and dynamics of two-compartment models. By harnessing the power of eigen analysis, we can gain insights into the long-term behaviour of the system, including equilibrium states and response to perturbations. Moreover, we explore the role of optimization methods in estimating parameters that define these models. Optimization techniques, such as gradient descent or genetic algorithms, are employed to fine-tune the model parameters, ensuring a more precise representation of the real-world system.

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We will dissect the mathematical foundations of these models, including the derivation of the governing equations and the application of eigen analysis, alongside practical examples to highlight their application in predicting system behaviour. The discussion will encompass strategies for parameter estimation through optimization, reflecting on how these methods enhance the model's predictive power. Furthermore, the paper will address the inherent challenges and limitations of two-compartment models, providing a comprehensive overview of their applications and significance in the broader context of mathematical modelling.

Methodology:

Two compartment model:

The model typically involves differential equations to describe the rate of transfer of the substance between the compartments. These rates can depend on various factors such as

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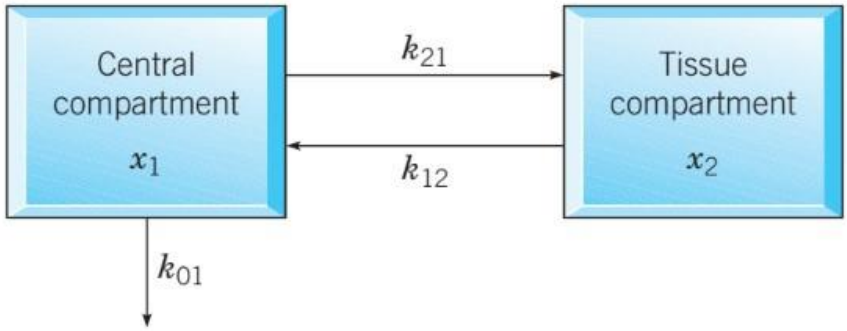
the amount of substance in each compartment and specific transfer rates, which are often key parameters in the model.



$$x_1 = -(k_{01} + k_{21})x_1 + k_{12}x_2$$

$$X_2 = +k_{21}x_1 + k_{12}x_2$$

Exponential peeling:

Exponential peeling involves iteratively removing, or "peeling off," data points from the dataset that are least consistent with an exponential model. By focusing on the subset of the data that best fits an exponential curve, the method seeks to enhance the accuracy of parameter estimation. This method is particularly relevant when analysing data sets that exhibit rapid changes initially, followed by a plateau or steady state, as is often the case in pharmacokinetic and other biological systems.

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Parameter Optimization (SSE) :

SSE stands for Sum of Squared Errors, a term commonly used in statistical analysis and is a measure of the total deviation of the response values from the fit provided by the model. if you have a set of observed values and a corresponding set of predicted values obtained from a model, the SSE is calculated as:

$$\text{SSE}= \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Here, n represents the number of data points $(y_i - \hat{y}_i)^2$ and represents the squared difference between the observed and predicted values, indicating the error for each individual observation.

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Eigenvalues / Eigenvectors:

An eigenvalue, typically denoted as λ, is a scalar that indicates the factor by which an eigenvector is scaled during a linear transformation. Mathematically, if A is a square matrix, an eigenvalue λ is a solution to the characteristic equation: det (A- λI) = 0. An eigenvector is a non-zero vector that changes by only a scalar factor when a linear transformation is applied to it. In other words, it's a vector that, when multiplied by matrix A, yields a simple scale of itself. For a given eigenvalue λ, an eigenvector v satisfies: = Av=λv. In systems of differential equations, eigenvalues can be used to analyse the stability of equilibrium points. The sign and magnitude of the eigenvalues give insights into the behaviour of the system near these points.

Part 1:

Part A

Given that K has been defined for the rates

$$K = \begin{bmatrix} K_{11} & K_{12} \\ K_{21} & K_{22} \end{bmatrix}$$

In order to prove the eigenvalues are real, distinct and negative, the determinant is given as det (K- λI) = 0 where I is the identity matrix as described as,

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Therefore

$$\text{Det}\begin{pmatrix} K_{11} - \lambda & K_{12} \\ K_{21} & K_{22} - \lambda \end{pmatrix}$$

$$= \det\begin{pmatrix} -(k_{01} + k_{21}) - \lambda & k_{21} \\ k_{12} & -k_{12} - \lambda \end{pmatrix} = 0$$

$$=> (-(k_{01} + k_{21}) - \lambda)(-k_{12} - \lambda) - (k_{21} * k_{12}) = 0$$

$$\Rightarrow \lambda^2 + k_{01}k_{12} + k_{21}k_{12} + \lambda k_{12} + (k_{01} + k_{21})\lambda - k_{21}k_{12} = 0$$

$$\Rightarrow \lambda^2 + (k_{01} + k_{21} + k_{12})\lambda + k_{01}k_{12} = 0$$

As per the methodology, the K matrix has real eigenvalues. In order to be distinct and negative the discriminant must be greater than 0 as per

$$b^2 - 4ac > 0$$

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Where a =1 , b = $(k_{01} + k_{21} + k_{12})$ and c = $k_{01}k_{12}$. Therefore

$$(k_{01} + k_{21} + k_{12})^2 - 4 * 1 * k_{01}k_{12} > 0$$

$$=> k_{01}^2 + k_{21}^2 + k_{12}^2 + k_{01}k_{12} + k_{01}k_{21} + k_{21}k_{12} - 4k_{01}k_{12} > 0$$

Since the polynomial satisfies the discriminant thus proves to have distinct values. As per the following computation

$$\Rightarrow \lambda_{1,2} = \frac{-(k_{01}+k_{21}+k_{12}) \pm \sqrt{(k_{01}+k_{21}+k_{12})^2 - 4*1*k_{01}k_{12}}}{2}$$

As per the previous discriminant proof you can assume that

$$\Rightarrow (k_{01} + k_{21} + k_{12})^2 > (k_{01} + k_{21} + k_{12})^2 - 4 * 1 * k_{01}k_{12}$$

$$\Rightarrow (k_{01} + k_{21} + k_{12}) > \sqrt{(k_{01} + k_{21} + k_{12})^2 - 4 * 1 * k_{01}k_{12}}$$

As the values are greater than the discriminant this proves that the eigenvalues are negative.

Part B:

In order to show that the eigenvalues $\lambda_1 + \lambda_2 = -(k_{01} + k_{21} + k_{12})$ and $\lambda_1\lambda_2 = k_{01}k_{12}$, implement the solutions previously attained to compute

$$\Rightarrow \lambda_1 = \frac{-(k_{01}+k_{21}+k_{12}) + \sqrt{(k_{01}+k_{21}+k_{12})^2 - 4*1*k_{01}k_{12}}}{2} \quad //$$

$$\Rightarrow \lambda_2 = \frac{-(k_{01}+k_{21}+k_{12}) - \sqrt{(k_{01}+k_{21}+k_{12})^2 - 4*1*k_{01}k_{12}}}{2}$$

$$\Rightarrow \lambda_1 + \lambda_2 = \frac{-2(k_{01}+k_{21}+k_{12})}{2}$$

$$\Rightarrow \lambda_1 + \lambda_2 = -(k_{01} + k_{21} + k_{12})$$

Thus proved $\lambda_1 + \lambda_2$. For $\lambda_1\lambda_2$,

$$=> \lambda_1\lambda_2 = \frac{(k_{01} + k_{21} + k_{12})^2 - (k_{01} + k_{21} + k_{12})^2 + 4k_{01}k_{12}}{4}$$

$$=> \frac{4k_{01}k_{12}}{4} = k_{01}k_{12}$$

Thus proved for $\lambda_1\lambda_2$.

Question 2

Part A: Estimating Eigenvalues and Eigenvectors of K from Transient Concentration Data

For the eigenvectors V_1 and V_2 that satisfy K as denoted by $\begin{pmatrix} v_{11} \\ v_{12} \end{pmatrix}$ and $\begin{pmatrix} v_{21} \\ v_{22} \end{pmatrix}$

where the solution for the above given system of equations is given as

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$$x(t) = \alpha e^{\lambda_1 t} v_1 + \beta e^{\lambda_2 t} v_2$$

For α and β which are defined as nonzero where $e^{(\lambda_1+\lambda_2)t} \sim 0$. With the acquired equation, x(t) can be rewritten as

$$x_1(t) \approx \alpha e^{\lambda_1 t} v_1 \text{ or } x(t)\ \alpha e^{\lambda_1 t} v_{11}$$

Taking the log on both sides,

$$\ln(x_1(t)) = \ln(\alpha e^{\lambda_1 t} v_1) = \ln(a\, v_{11}) + \lambda_1 t$$

As per the same logic,

$$x_2(t) \approx \alpha e^{\lambda_2 t} v_2$$

$$\ln(x_2(t)) = \ln(\alpha e^{\lambda_2 t} v_2) = \ln(a\, v_{21}) + \lambda_2 t$$

As seen in the log functions, they appear as linear functions that can be plotted on the same graph and due to sharing the same slope, the intercepts $\ln(a\, v_{11})$ and $\ln(a\, v_{21})$ can be approximately straight lines.

Part B:

Using the original x(t) equation as derived by:

$$x(t) = e^{\lambda_1 t}(a v_1 + \beta e^{(\lambda_2-\lambda_1)t} v_2)$$

Where $(\lambda_2 - \lambda_1)t \approx 0$,the residual equation for the above solution is given as

$$x^*{}_r(t_n) = x^*(t_n) - \hat{v}_1^{(a)} e^{\widehat{\lambda_1} t_n}$$

For the estimates of $\lambda_2, \beta v_{12}$ and βv_{22}. Rearranging the original equation,

$$x^*(t_n) = x^*{}_r(t_n) + \hat{v}_1^{(a)} e^{\widehat{\lambda_1} t_n}$$

Applying the equation to the residual as follows:

$$x(t_n) + (a)t_n = ae + \beta e^{\sqrt{8}}$$

$$x + t_n = \alpha e^{v_1} + \beta e^{2\sqrt{2}} - \alpha(t)$$

$$x + t_n = \alpha e^{11} v_1 + \frac{\beta e^{d12} t_n}{\sqrt{2}} - \sqrt{\alpha} d t_n => x^*(t_n) \approx \beta e^{\lambda_2 t} v_2$$

Taking the log of both sides,

$$x_1(t_n) = \beta e^{\lambda_2} v_{12} \qquad x_2(t_n) = \beta e^{\lambda_2} v_{22}$$

$$\ln(x(t_n)) \approx \ln(\beta e^{\lambda_2 t} v_2)$$

$$\ln(x_1(t_n)) \approx \ln(\beta v_{12}) + \lambda_2 t \qquad \ln(x_2(t_n)) \approx \ln(\beta v_{22}) + \lambda_2 t$$

Thus proved.

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Part 3:

Given that the eigenvalues of $\lambda_1$and λ_2 satisfy the solution for the system of equations presented in the previous problem, you need to use the determinant (K – λI) = 0 to prove the system of equations given as :

$$\begin{pmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{pmatrix} \begin{pmatrix} K_{11} \\ K_{12} \end{pmatrix} = \begin{pmatrix} \lambda_1 v_{11} \\ \lambda_2 v_{12} \end{pmatrix}$$

$$\begin{pmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{pmatrix} \begin{pmatrix} K_{21} \\ K_{22} \end{pmatrix} = \begin{pmatrix} \lambda_1 v_{21} \\ \lambda_2 v_{22} \end{pmatrix}$$

As well as implementing the solution by K V=V Λ through:

$$V = \begin{pmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{pmatrix} \text{ and } \lambda = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$$

Lets assume that A matrix shares the same column space as K and can be written as

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

Using the determinant formula on the K matrix:

$$K = \begin{pmatrix} K_{11} & K_{12} \\ K_{21} & K_{22} \end{pmatrix}$$

(K – A_1 * I) (V) = 0

$$\begin{pmatrix} K_{11} - a_1 & K_{12} \\ K_{21} & K_{22} - a_1 \end{pmatrix} \begin{pmatrix} v_{11} \\ v_{21} \end{pmatrix} = 0$$

(1) $(K_{11} - a_1)v_{11} + K_{12}v_{21} = 0$

(2) $K_{21}v_{11} + (K_{22} - a_1)v_{21} = 0$

⇨ $K_{11}v_{11} + K_{12}v_{21} = a_1 v_{11}$ from (1)

⇨ $K_{21}v_{11} + K_{22}v_{21} = a_1 v_{11}$ from (2)

(K – A_2 * I) (V) = 0

$$\begin{pmatrix} K_{11} - a_2 & K_{12} \\ K_{21} & K_{22} - a_2 \end{pmatrix} \begin{pmatrix} v_{11} \\ v_{21} \end{pmatrix} = 0$$

(3) $(K_{11} - a_2)v_{11} + K_{12}v_{21} = 0$

(4) $K_{12}v_{11} + (K_{11} - a_2)v_{21} = 0$

⇨ $K_{11}v_{11} + K_{12}v_{21} = a_2 v_{11}$ from (1)

⇨ $K_{21}v_{11} + K_{22}v_{21} = a_2 v_{11}$ from (2)

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As per the derived equations the eigenvectors that prove the system of equations as derived by:

$$K_{11}v_{11} + K_{12}v_{21} = a_1v_{11}$$
$$K_{21}v_{11} + K_{22}v_{21} = a_1v_{11}$$
$$K_{11}v_{11} + K_{12}v_{21} = a_2v_{11}$$
$$K_{21}v_{11} + K_{22}v_{21} = a_2v_{11}$$

Part 4:

Using the equations derived from the estimation of λ_1+λ_2, we can derive k_{01}through

$$\Rightarrow \lambda_1 + \lambda_2 = -(k_{01} + k_{21} + k_{12})$$
$$\Rightarrow \lambda_1\lambda_2 = k_{01}k_{12}$$

Let's assume that a $\widehat{k_{01}}$ exists such it satisfies the above equations. To prove that we can determine

$$\Rightarrow \hat{\lambda}_1 + \hat{\lambda}_2 = -(\hat{k}_{01} + \hat{k}_{21} + \hat{k}_{12})$$
$$\Rightarrow \hat{k}_{01} = -(\hat{\lambda}_1 + \hat{\lambda}_2) + \hat{k}_{21} + \hat{k}_{12}$$

This can be also derived through $\lambda_1\lambda_2$ as

$$\hat{\lambda}_1\hat{\lambda}_2 = \hat{k}_{01}\hat{k}_{12}$$
$$\hat{k}_{01} = \frac{\hat{\lambda}_1\hat{\lambda}_2}{\hat{k}_{12}}$$

Thus it can be proved.

Part 5:

As per the previous problems the system of equations could be mapped linearly through the log of the solution as ln x_1(t) and ln x_2 (t) with slopes equal to λ_1 and ln(a_1v_{11}) and ln(a_1v_{21}) as the intercepts which is given as:

$$y_1 = \ln(x_1t) = -0.0161t - 1.633$$
$$y_2 = \ln(x_2t) = -0.0161t - 1.474$$

Where $\lambda_1 = -0.0161$, ln(av_{11}) = -1.633 and ln(av_{21}) = -1.474. Using the same logic, you can form the equations of the estimate from question 2b that is approximately:

$$y_1^* = \ln(x_1^*t) = -0.07399t - 1.5$$
$$y_2^* = \ln(x_2^*t) = -0.07399t - 3.8$$

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Where $a_2 = -0.07399$, ln(bv_{12}) = -1.5 and ln(bv_{12}) = -3.8. Using the equations, you have learnt the

$$V_1 = \begin{pmatrix} 0.1874 & 0.2293 \\ 0.02006 & -0.01594 \end{pmatrix} / V_2 = \begin{pmatrix} 0.1874 & 0.2293 \\ 0.02006 & 0.2293 \end{pmatrix}$$

As eigenvectors of the K matrix, the eigenvalues corresponding to those vectors, $\lambda_1 = -0.0161$ and $\lambda_2 = -0.07399$ produce the rates for the system of equation as given by:

$$\begin{pmatrix} -0.1533 & 0.0068 \\ 0.1 & 0.1 \end{pmatrix}$$

As substituted to find the original values is:

$$K = \begin{bmatrix} -k_{01} - k_{21} & k_{12} \\ k_{21} & -k_{12} \end{bmatrix}$$

Where $k_{01} = 0.0533$, $k_{12} = 0.0068$ and $k_{21} = 0.1$.

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Appendix:

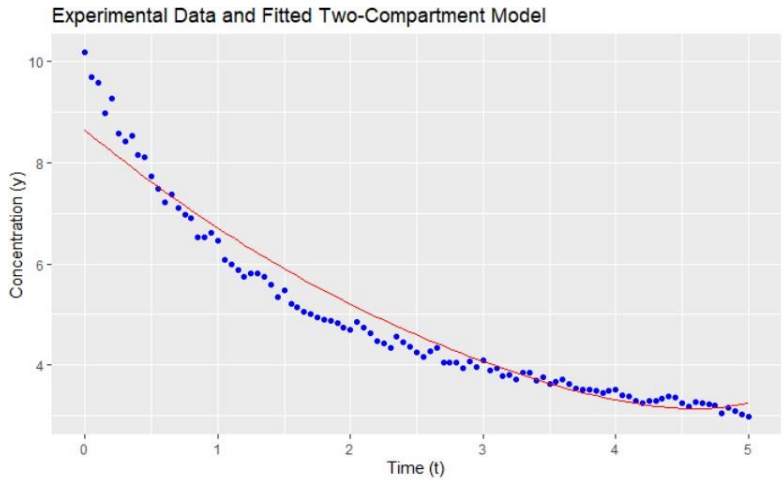

Figure 1(a) Exponential peeling

The figure above is a graphical representation of applying exponential peeling to the estimated parameters and provides an approximately equal fit to the data points.

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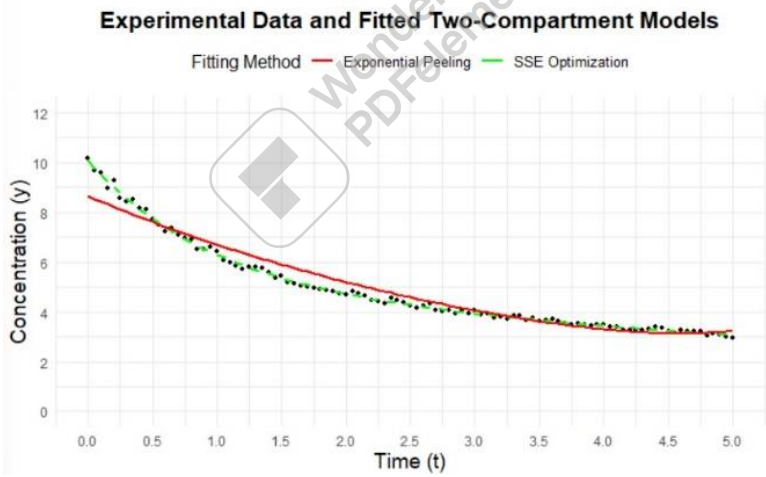

Figure 1(b) SSE Optimization

The figure above is a graphical representation of minimizing the residuals to fit the plotted data points compared to the exponential peeling where the SSE outperforms fitted model as it fits all points more accurately.

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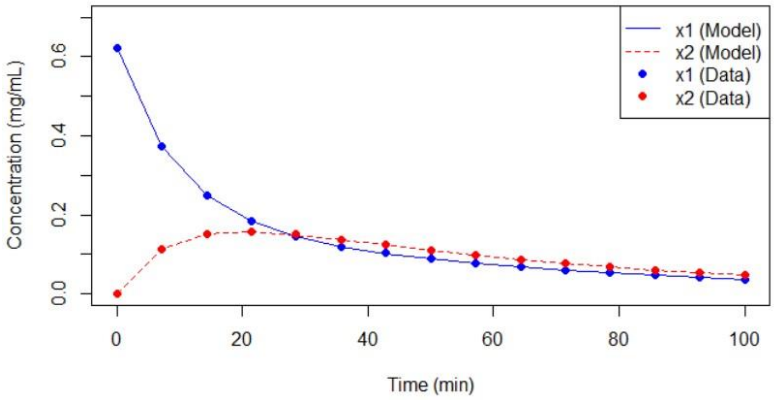

Figure 2 Solution for System of equations

The figure above provides a graphical presentation of log of exponential functions $x_1(t)$ and $x_2(t)$ depicting the models following a nearly linear path proving that the solution derived is accurate.

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```
---
title: "Part.1.A"
output: html_document
date: "2023-12-01"
---

```{r Library, include=FALSE}
knitr::opts_chunk$set(echo = TRUE)

library(dplyr)

## 'dplyr'contains the 'package:stats': ##
##     filter, lag
## 'package:base': includes the following
##     intersect, setdiff, setequal, union

library(ggplot2)
library(deSolve)
library("rlang")
library("reticulate") library(minpack.lm)
library(deSolve)
# Use sympy for symbolic mathematics
sympy <- import("sympy")

```

R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

```{r Part 1.A}
# load the data from the global environment
data <- project.data

# Start by fitting a single exponential function
single_exp_fit <- nls(data$y ~ b1 * exp(b2 * data$time), data = data, start = list(b1 = 1, b2 =
1))

# Capture the first set of estimated parameters
b1_estimate <- coef(single_exp_fit)["b1"]
b2_estimate <- coef(single_exp_fit)["b2"]

# Next, subtract the contribution of the first exponential from the data
data_residual <- data %>%
  mutate(residual = y - b1_estimate * exp(b2_estimate * time))

# Fit a single exponential to the residual
single_exp_residual_fit <- nls(residual ~ b3 * exp(b4 * data$time), data = data_residual, start
= list(b3 = 1))

# Capture the second set of estimated parameters
b3_estimate <- coef(single_exp_residual_fit)["b3"]
b4_estimate <- coef(single_exp_residual_fit)["b4"]
# Now you have the estimated parameters b1, b2, b3, and b4 from the exponential peeling process

# Print the estimated parameters
print(c(b1_estimate, b2_estimate, b3_estimate, b4_estimate))

# Define the two-compartment model using the estimated parameters
two_compartment_estimated <- function(t) {
  return(b1_estimate * exp(b2_estimate * t) + b3_estimate * exp(b4_estimate * t)) }
```

Including Plots
```

You can also embed plots, for example:

```{r Part 1.B, echo=FALSE}
# Create a data frame for plotting the estimated model
plot_data <- data %>%
mutate(estimated_model = two_compartment_estimated(time))

# Create a plot
ggplot(data) +
  geom_point(aes(x = time, y = y), color = "blue") +
  geom_line(data = plot_data, aes(x = time, y = estimated_model), color = "red") + labs(x =
"Time (t)" ggtitle("Experimental Data and Fitted Two-Compartment Model")

```



```{r Bonus, include=FALSE}

# Define the two-compartment model
two_compartment <- function(params, t) {
  b1 <- params[1]
  b2 <- params[2]
  b3 <- params[3]
  b4 <- params[4]
  return(b1 * exp(b2 * t) + b3 * exp(b4 * t))
}

# Define the SSE objective function
sse_objective <- function(params) {
  predicted <- two_compartment(params, data$time)
  SSE <- sum((data$y - predicted)^2)
  return(SSE)
}

initial_params <- c(b1 = b1_estimate, b2 = b2_estimate, b3 = b3_estimate, b4 = b4_estimate)

# Minimize the SSE using the optim function
result <- optim(par = initial_params, fn = sse_objective, method = "L-BFGS-B")

# Extract the estimated parameters
estimated_params_sse <- result$par

# Create a data frame for plotting the models
plot_data <- data %>%
mutate(estimated_model_sse = two_compartment(estimated_params_sse, time),
# Add estimated_model_sse estimated_model = two_compartment(initial_params, time))

# Add estimated_model

ggplot(plot_data) +
  geom_point(aes(x = time, y = y), color = "black", size = 1) +
  geom_line(aes(x = time, y = estimated_model_sse, color = "SSE Optimization"), linetype =
"dashed", size = geom_line(aes(x = time, y = estimated_model, color = "Exponential Peeling"),
size = 1) +
labs(x = "Time (t)", y = "Concentration (y)") + ggtitle("Experimental Data and Fitted Two-
Compartment Models") + theme_minimal() + theme(legend.position = "top",legend.title =
element_text(size = 12),legend.text = element_text(size = 10),
plot.title = element_text(size = 16, hjust = 0.5, face = "bold"),
axis.title.x = element_text(size = 14),
axis.title.y = element_text(size = 14)) +
scale_x_continuous(limits = c(0, 5), breaks = seq(0, 5, by = 0.5)) +
scale_y_continuous(limits = c(0, 12), breaks = seq(0, 12, by = 2)) +
scale_color_manual(values = c("SSE Optimization" = "green", "Exponential Peeling" = "red")) +
labs(color = "Fitting Method")

```

```
Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R
code that generated the plot.
```

```
---
title: "R Notebook"
output: html_notebook
---

This is an [R Markdown](http://rmarkdown.rstudio.com) Notebook. When you execute code within
the notebook, the results appear beneath the code.

Try executing this chunk by clicking the *Run* button within the chunk or by placing your
cursor inside it and pressing *Ctrl+Shift+Enter*.

```{r}
# loaded data from the global environment
time <- math.486.project.part.2.5$time
x1 <- math.486.project.part.2.5$x1
x2 <- math.486.project.part.2.5$x2

estimator <- function(t, vec, para) {
    with(as.list(c(vec, para)), {
        # Assign  the rates of change for each state variable
        # Assuming state variables are x and y, and parameters are a, b, c, and d
        x1 <- - (a + b) * x1 + c * x2
        x2 <- b * x1 - d * x2

        # Return a list containing the rates of change
        return(list(c(x1,x2)))
    })
}

init_cond <-  c(x1 = 0.623, x2 = 0.0)

residual_val <- function(ob, time, x1, x2) {
  obj <- as.list(ob)
  solution <- ode(y = init_cond, times = time, func = estimator, parms = obj)
  coef_sol <- solution[, c("x1", "x2")]
  obs_values <- cbind(x1, x2)
  residual_vals <- as.vector(model_values - obs_values)
  return(residuals)
}

# Fit the model to the data
fit <- nls.lm(par = c(k01 = 0.1, k12 = 0.3, k21 = 0.2), fn = residual_val, time = time, x1 =
x1, x2 = x2)

# Extract estimated parameters
estimated_params <- coef(fit)

cat("Estimated Parameter Values:\n")
cat("k01:", estimated_params["k01"], "\n")
cat("k12:", estimated_params["k12"], "\n")
cat("k21:", estimated_params["k21"], "\n")

# Solve the system with the estimated parameters
estimated_sol <- ode(y = init_cond, times = time, func = estimator, parms = estimated_params)

# Plot the estimated solution and data with the same set of coordinates on the axes
plot(time, estimated_sol[, "x1"], type = "l", col = "blue", lty = 1, xlab = "Time (min)", ylab
= "Concentration (mg/mL)", lines(time, estimated_sol[, "x2"], type = "l", col = "red", lty =
2))
points(time, x1, col = "blue", pch = 16)
points(time, x2, col = "red", pch = 16)

# Add legend
legend("topright", legend = c("x1 (Model)", "x2 (Model)", "x1 (Data)", "x2 (Data)"), col =
c("blue", "red"))
```

Add a new chunk by clicking the *Insert Chunk* button on the toolbar or by pressing
Ctrl+Alt+I.
```

```
When you save the notebook, an HTML file containing the code and output will be saved alongside
it (click the *Preview* button or press *Ctrl+Shift+K* to preview the HTML file).

The preview shows you a rendered HTML copy of the contents of the editor. Consequently, unlike
*Knit*, *Preview* does not run any R code chunks. Instead, the output of the chunk when it was
last run in the editor is displayed.
```

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Bibliography:

Holmes, Mark . "Traffic Flow." *Introduction to the Foundations of Applied Mathematics*, vol. 56, no. 0939-2475, Mar. 2009, pub.math.leidenuniv.nl/~chirilusbrucknerm/ds_seminar_2016/holmes_Ch5_traffic_flow.pdf, https://doi.org/10.1007/978-0-387-87765-5. Accessed 16 Dec. 2021.

Grolemund, Garrett. *D Loading and Saving Data in R | Hands-on Programming with R. Rstudio-Education.github.io*, O'Reilly , rstudio-education.github.io/hopr/dataio.html.

Wickham, Hadley . "Function Reference." *Ggplot2.Tidyverse.org*, ggplot2.tidyverse.org/reference/.

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